ESP Docs AI Assistant

MCP Server

This custom large language model (LLM), trained on official documentation from espressif.com, is designed to provide technical support and answers related to Espressif's products and services. Give it a try, share your thoughts, and let us know your feedback—we truly appreciate it!

Note: AI-generated information may be incomplete or inaccurate. Always verify critical information with official sources.

> I am very very interested in using esp-dsp, but I'm stuck on a very important thing that I forgot: I have both a transfer function and difference equation for a low-pass filter, but I need to figure out what the coefficents should before I can invoke esp-dsp filter modules. Can you jar my memory as to how derive the filter coefficients based on the available information i described?

The knowledge sources do not provide the step-by-step mathematical theory for converting an arbitrary continuous transfer function to discrete coefficients. However, they explicitly define the coefficient structure and conventions expected by the **esp-dsp** filter modules Esp-dsp API Reference , as well as built-in functions to calculate them directly.

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1. How Coefficients Map to Difference Equations in esp-dsp

For standard IIR (biquad) filters, the standard difference equation relates input x[n] and output y[n]:

```
y[n] = b0*x[n] + b1*x[n-1] + b2*x[n-2] - a1*y[n-1] - a2*y[n-2]
```

Corresponding to the discrete transfer function:

```
H(z) = (b0 + b1*z^(-1) + b2*z^(-2)) / (1 + a1*z^(-1) + a2*z^(-2))
```

In the **esp-dsp** library:

Ask a follow-up

\+ Fast

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